UNIVERSITY OF THE PUNJAB

B.S. 4 Years Program : First Semester – Fall 2021

Roll No. ....

Subject: Calculus and Analytical Geometry **Paper:** MS-152 A **Time:** 3 Hrs. **Marks:** 60

Q.1. Solve the following questions. (6x5=30)

(i) Find the limit $\lim_{h \to 0} \frac{h^2}{1 - \cos h}$.

(ii) Given $x^3 + y^3 = 3xy^2$. Find $\frac{dy}{dx}$ by implicit differentiation.

(iii) Let $y = (\ln x)^{\tan x}$. Find $\frac{dy}{dx}$.

(iv) Evaluate the integral $\int \frac{e^{\tan^{-1} x}}{1 + x^2} dx$.

(v) Evaluate the integral $\int \sin^4 x \cos^3 x dx$.

(vi) Find an equation of the plane that passes through the points $A(-2,1,1)$, $B(0,2,3)$, and $C(1,0,-1)$.

Solve the following questions. (3x10=30)

Q.2. Let $f(x) = x^4 - 12x^3$. Find the relative extrema using both first and second derivative tests.

Q.3. Evaluate the improper integral $\int_{-2}^{2} \frac{dx}{x^2} dx$ if it converges.

Q.4. Show that the lines $L_1 : x = 2 - t,\ \ y = 2t,\ \ z = 1 + t$ and $L_2 : x = 1 + 2t,\ \ y = 3 - 4t,\ \ z = 5 - 2t$ are parallel, and find the distance between them.

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i) $\lim_{h\to 0} \frac{h^2}{1-\cosh}$ $\left(\frac{0}{0}\right)$

By L.hospital Rule

$$= \lim_{h\to 0}\left(\frac{\frac{d}{dx}h^2}{\frac{d}{dx}(1-\cosh)}\right)$$

$$= \lim_{h\to 0} \frac{2h}{\sinh} \quad \left(\frac{0}{0}\right)$$

Again L.H.R

$$= \lim_{h\to 0} \frac{2}{\cosh}$$

$$= \frac{2}{1} \Rightarrow 2$$

ii) Given $x^3+y^3=3xy^2$. Find $\frac{dy}{dx}$ by implicit differentiation.

diff w.r.t 'x'

$$3x^2+3y^2\frac{dy}{dx} = 3x\cdot 2y\frac{dy}{dx}+3y^2$$

$$3y^2\frac{dy}{dx}-6xy\frac{dy}{dx} = 3y^2-3x^2$$

$$\frac{dy}{dx}\left(3y^2-6xy\right) = 3(y^2-x^2)$$

$$\frac{dy}{dx} = \frac{\cancel{3}(y^2-x^2)}{\cancel{3}(y^2-2xy)}$$

$$\frac{dy}{dx} = \frac{y^2-x^2}{y^2-2xy}$$

iii) Let $y=(\ln x)^{\tan x}$. Find $\frac{dy}{dx}$.

$$\ln y = \tan x \ln(\ln x)$$

diff w.r.t 'x'

$$\frac{1}{y}\frac{dy}{dx} = \sec^2 x \ln(\ln x) + \tan x\cdot\frac{1}{\ln x}\cdot\frac{1}{x}$$

$$\frac{dy}{dx} = y\left(\sec^2 x\cdot\ln(\ln x)+\frac{\tan x}{x\ln x}\right)$$

$$= (\ln x)^{\tan x}\left(\sec^2 x\cdot\ln(\ln x)+\frac{\tan x}{x\ln x}\right)$$

iv) Evaluate the integral $\int \frac{e^{\tan^{-1}x}}{1+x^2} dx$.

Let $\tan^{-1}x = \theta$

$x = \tan\theta$

$dx = \sec^2\theta \, d\theta$

$\frac{1}{1+x^2} = \cos^2\theta$

$\frac{d}{dx}(\tan^{-1}x) = \frac{1}{1+x^2}$

$\int e^{\theta} \cos^2\theta \sec^2\theta \, d\theta$

$\because \sec^2\theta = 1+\tan^2\theta$

$\int e^{\theta}\cos^2\theta(1+\tan^2\theta)\,d\theta$

$\int e^{\theta}(\cos^2\theta + \cos^2\theta\tan^2\theta)\,d\theta$

$= \int e^{\theta}\left(\cos^2\theta + \cancel{\cos^2\theta}\,\frac{\sin^2\theta}{\cancel{\cos^2\theta}}\right)d\theta$

$\because \cos^2\theta + \sin^2\theta = 1$

$= \int e^{\theta} \cdot 1 \, d\theta$

$= e^{\theta} + C$

$= e^{\tan^{-1}x} + C$

v) Evaluate the integral $\int \sin^4 x \cos^3 x \, dx$.

$= \int \sin^4 x \cdot \cos^2 x \cdot \cos x \, dx$

$\because \sin x = u$

$\cos x\,dx = du$

$= \int \sin^4 x(1-\sin^2 x)\cos x \, dx$

$= \int u^4(1-u^2)\,du$

$= \int (u^4 - u^6)\,du$

$= \int u^4 du - \int u^6 du$

$= \frac{u^5}{5} - \frac{u^7}{7} + C$

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$= \frac{\sin^5 x}{5} - \frac{\sin^7 x}{7} + C$

vi) Find an equation of the plane that passes through the points $A(-2,1,1)$, $B(0,2,3)$, and $C(1,0,-1)$.

(I)

find vectors lie on the plane

$\vec{AB} = B - A \Rightarrow (0,2,3) - (-2,1,1)$

$= (2,1,2)$

$\vec{AC} = C - A \Rightarrow (1,0,-1) - (-2,1,1)$

(II) $= (3,-1,-2)$

find $\vec{n}$

$\vec{n} = \vec{AB} \times \vec{AC}$

$$= \begin{vmatrix} i & j & k \\ 2 & 1 & 2 \\ 3 & -1 & -2 \end{vmatrix}$$

$$= i\begin{vmatrix} 1 & 2 \\ -1 & -2 \end{vmatrix} - j\begin{vmatrix} 2 & 2 \\ 3 & -2 \end{vmatrix} + k\begin{vmatrix} 2 & 1 \\ 3 & -1 \end{vmatrix}$$

$= i(-2+2) - j(-4-6) + k(-2-3)$

$\vec{n} \Rightarrow 10j - 5k$

$\vec{n} = (0, 10, -5)$

(III) equation of plane

$Ax + By + Cz = D$

$0x + 10y - 5z = D$

Subtitute coordinate point of $A(-2,1,1)$ into eq

$0(-2) + 10(1) - 5(1) = D$

$10 - 5 = D$

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$D = 5$

CamScanner

$10y - 5z = 5$

$\boxed{2y - z = 1}$

Long Questions

Let $f(x) = x^4 - 12x^3$. Find the relative extrema using both first and second derivative tests.

By second derivative:

$$f(x) = x^4 - 12x^3$$

$$f'(x) = 4x^3 - 36x^2$$

$$4x^3 - 36x^2 = 0$$

$$x^2(4x - 36) = 0$$

$$4x - 36 = 0 \quad , \quad x^2 = 0$$

$$4x = 36$$

$$x = 9 \quad , \quad x = 0$$

At $x = 9$

$$f''(x) = 12x^2 - 72x$$

$$f''(9) = 12(9)^2 - 72(9)$$

$$= 324 \quad \text{(relative minima)}$$

use first derivative to test at $x = 0$

$$f'(x) = 4x^3 - 36x^2$$

$$f'(0 - \varepsilon) = 4(0 - \varepsilon)^3 - 36(0 - \varepsilon)^2$$

$$= -4\varepsilon^3 - 36\varepsilon$$

$$=$$

$$f'(0+\varepsilon) = 4(0+\varepsilon)^3 - 36(0+\varepsilon)^2 \Rightarrow 4\varepsilon^3 - 36\varepsilon^2$$

relative minima

3. Evaluate the improper integral $\int_{-2}^{2} \frac{dx}{x^2}\, dx$ if it converges.

$$\int_{-2}^{2} \frac{dx}{x^2} = \int_{-2}^{0} \frac{dx}{x^2} + \int_{0}^{2} \frac{dx}{x^2}$$

$$= \left| \frac{x^{-2+1}}{-2+1} \right|_{-2}^{0} + \left| \frac{x^{-2+1}}{-2+1} \right|_{0}^{2}$$

$$= \left| -\frac{1}{x} \right|_{-2}^{0} + \left| \frac{-1}{x} \right|_{0}^{2}$$

$$= \left(\frac{-1}{0} + \frac{1}{(-2)} \right)$$

$= -\infty$ divergent.

$$= \left(\frac{-1}{2} + \frac{1}{0} \right)$$

$= \infty$ divergent

both parts of improper integral are divergent.